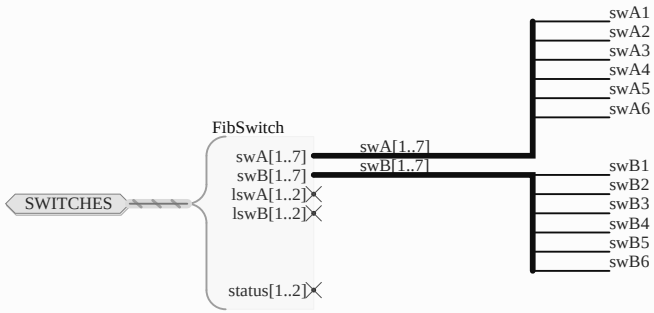
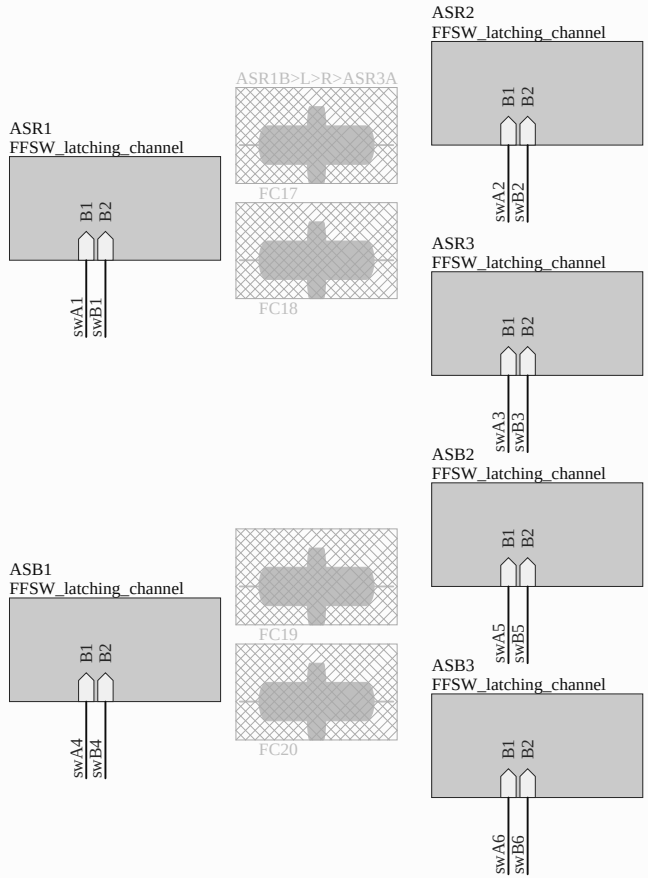




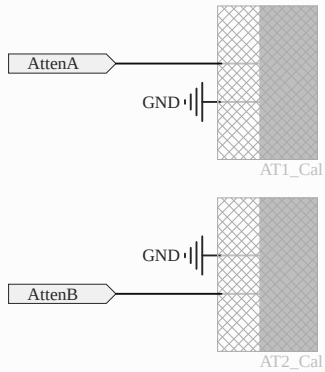


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File:	C:\Users\..\AS_Splitting.SchDoc	Drawn By:

FVOA:
Resistive load around 163 ohms per datasheet inference (3.5v @ 80mW). Lab test indicates more like 625 Ohms, 8mA at full atten. Response is ~loglinear in ~1 - 4.5 V window 1-50dB per test report.

Provision 10mA @ 5V per atten because agiltron is agiltron. 120mA for all 12 attenuators on a board.

DAC7578 8 channel, 3.3-5v needs opamps. Use a pair with alternate addresses to drive all 12 attenuators.



MillMax Pin Sockets

J8_Cal
0552-2-15-15-11-27-10-0

J10_Cal
0552-2-15-15-11-27-10-0

J9_Cal
0552-2-15-15-11-27-10-0

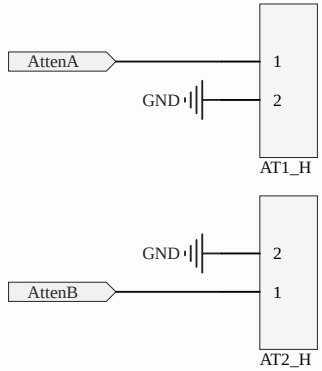
J11_Cal
0552-2-15-15-11-27-10-0

Title		
Size: A	Number	Revision
Date: 10/21/2026		Sheet of
File: C:\Users\..\atten_channel.SchDoc		Drawn By:

FVOA:
Resistive load around 163 ohms per datasheet inference (3.5v @ 80mW). Lab test indicates more like 625 Ohms, 8mA at full atten. Response is ~loglinear in ~1 - 4.5 V window 1-50dB per test report.

Provision 10mA @ 5V per atten because agiltron is agiltron. 120mA for all 12 attenuators on a board.

DAC7578 8 channel, 3.3-5v needs opamps. Use a pair with alternate addresses to drive all 12 attenuators.



MillMax Pin Sockets

J8_H
0552-2-15-15-11-27-10-0

J10_H
0552-2-15-15-11-27-10-0

J9_H
0552-2-15-15-11-27-10-0

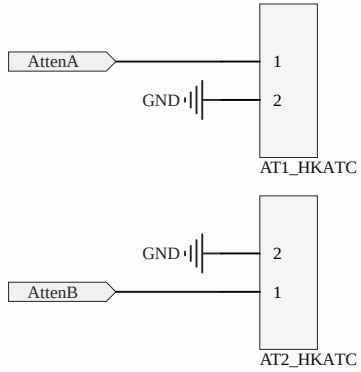
J11_H
0552-2-15-15-11-27-10-0

Title		
Size: A	Number	Revision
Date: 10/21/2026		Sheet of
File: C:\Users\..\atten_channel.SchDoc		Drawn By:

FVOA:
Resistive load around 163 ohms per datasheet inference (3.5v @ 80mW). Lab test indicates more like 625 Ohms, 8mA at full atten. Response is ~loglinear in ~1 - 4.5 V window 1-50dB per test report.

Provision 10mA @ 5V per atten because agiltron is agiltron. 120mA for all 12 attenuators on a board.

DAC7578 8 channel, 3.3-5v needs opamps. Use a pair with alternate addresses to drive all 12 attenuators.



MillMax Pin Sockets

J8_HKATC
0552-2-15-15-11-27-10-0

J10_HKATC
0552-2-15-15-11-27-10-0

J9_HKATC
0552-2-15-15-11-27-10-0

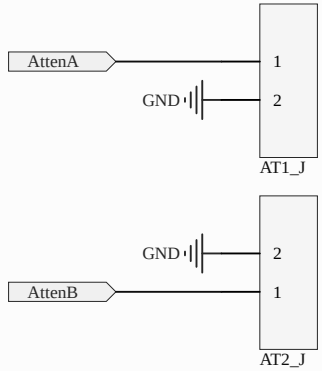
J11_HKATC
0552-2-15-15-11-27-10-0

Title		
Size: A	Number	Revision
Date: 10/21/2026		Sheet of
File: C:\Users\..\atten_channel.SchDoc		Drawn By:

FVOA:
Resistive load around 163 ohms per datasheet inference (3.5v @ 80mW). Lab test indicates more like 625 Ohms, 8mA at full atten. Response is ~loglinear in ~1 - 4.5 V window 1-50dB per test report.

Provision 10mA @ 5V per atten because agiltron is agiltron. 120mA for all 12 attenuators on a board.

DAC7578 8 channel, 3.3-5v needs opamps. Use a pair with alternate addresses to drive all 12 attenuators.



MillMax Pin Sockets

J8_J
0552-2-15-15-11-27-10-0

J10_J
0552-2-15-15-11-27-10-0

J9_J
0552-2-15-15-11-27-10-0

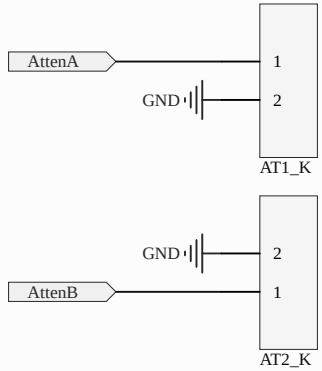
J11_J
0552-2-15-15-11-27-10-0

Title		
Size: A	Number	Revision
Date: 10/21/2026		Sheet of
File: C:\Users\..\atten_channel.SchDoc		Drawn By:

FVOA:
Resistive load around 163 ohms per datasheet inference (3.5v @ 80mW). Lab test indicates more like 625 Ohms, 8mA at full atten. Response is ~loglinear in ~1 - 4.5 V window 1-50dB per test report.

Provision 10mA @ 5V per atten because agiltron is agiltron. 120mA for all 12 attenuators on a board.

DAC7578 8 channel, 3.3-5v needs opamps. Use a pair with alternate addresses to drive all 12 attenuators.



MillMax Pin Sockets

J8_K
0552-2-15-15-11-27-10-0

J10_K
0552-2-15-15-11-27-10-0

J9_K
0552-2-15-15-11-27-10-0

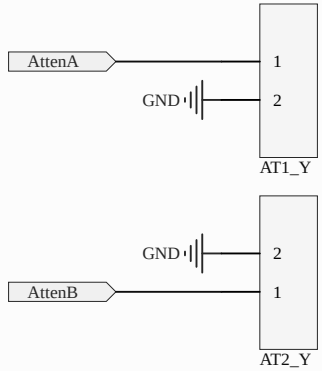
J11_K
0552-2-15-15-11-27-10-0

Title		
Size: A	Number	Revision
Date: 10/21/2026		Sheet of
File: C:\Users\..\atten_channel.SchDoc		Drawn By:

FVOA:
Resistive load around 163 ohms per datasheet inference (3.5v @ 80mW). Lab test indicates more like 625 Ohms, 8mA at full atten. Response is ~loglinear in ~1 - 4.5 V window 1-50dB per test report.

Provision 10mA @ 5V per atten because agiltron is agiltron. 120mA for all 12 attenuators on a board.

DAC7578 8 channel, 3.3-5v needs opamps. Use a pair with alternate addresses to drive all 12 attenuators.



MillMax Pin Sockets

J8_Y
0552-2-15-15-11-27-10-0

J10_Y
0552-2-15-15-11-27-10-0

J9_Y
0552-2-15-15-11-27-10-0

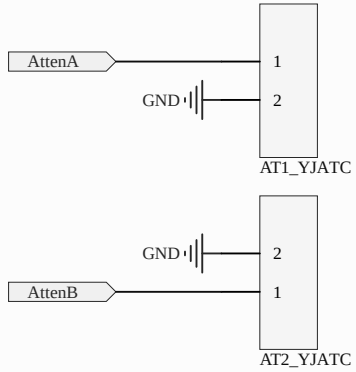
J11_Y
0552-2-15-15-11-27-10-0

Title		
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Date: 10/21/2026		Sheet of
File: C:\Users\..\atten_channel.SchDoc		Drawn By:

FVOA:
Resistive load around 163 ohms per datasheet inference (3.5v @ 80mW). Lab test indicates more like 625 Ohms, 8mA at full atten. Response is ~loglinear in ~1 - 4.5 V window 1-50dB per test report.

Provision 10mA @ 5V per atten because agiltron is agiltron. 120mA for all 12 attenuators on a board.

DAC7578 8 channel, 3.3-5v needs opamps. Use a pair with alternate addresses to drive all 12 attenuators.



MillMax Pin Sockets

J8_YJATC
0552-2-15-15-11-27-10-0

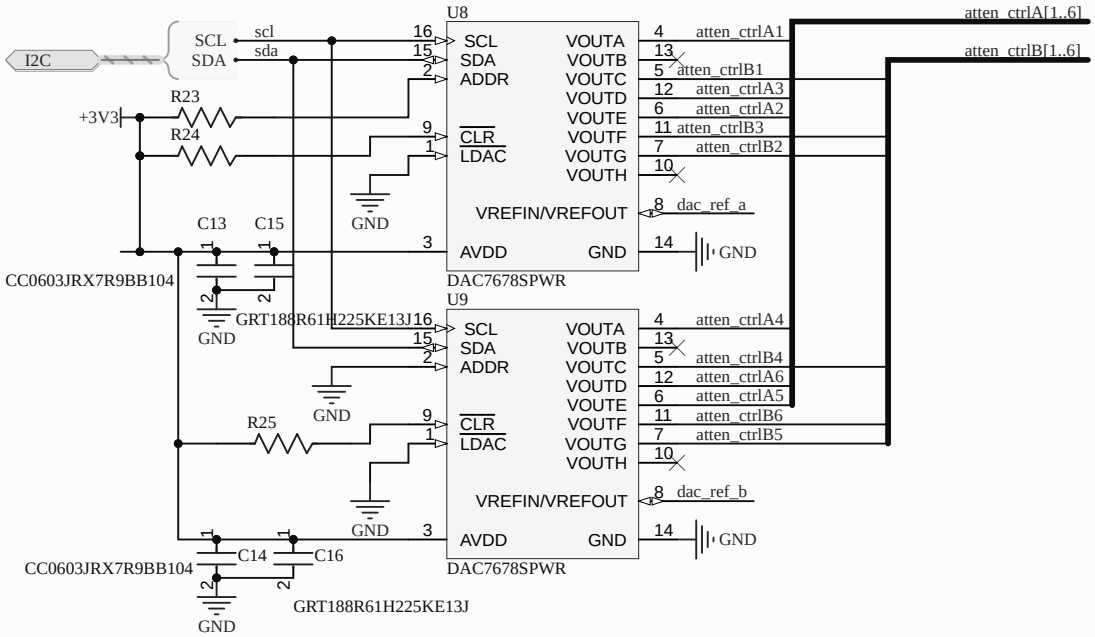
J10_YJATC
0552-2-15-15-11-27-10-0

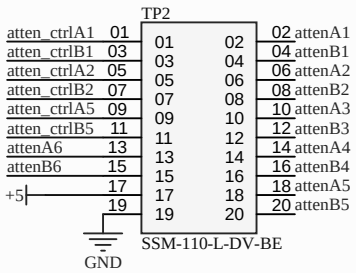
J9_YJATC
0552-2-15-15-11-27-10-0

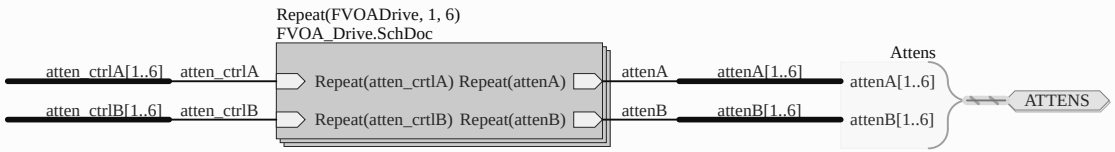
J11_YJATC
0552-2-15-15-11-27-10-0

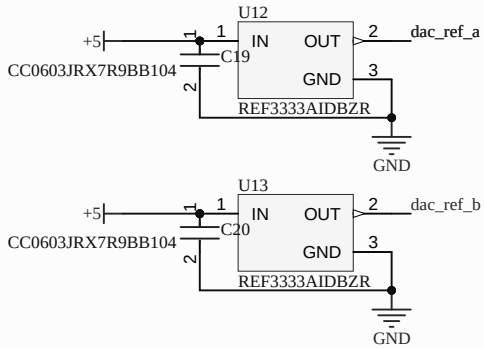
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REVISION	DESCRIPTION	DATE	APPROVED





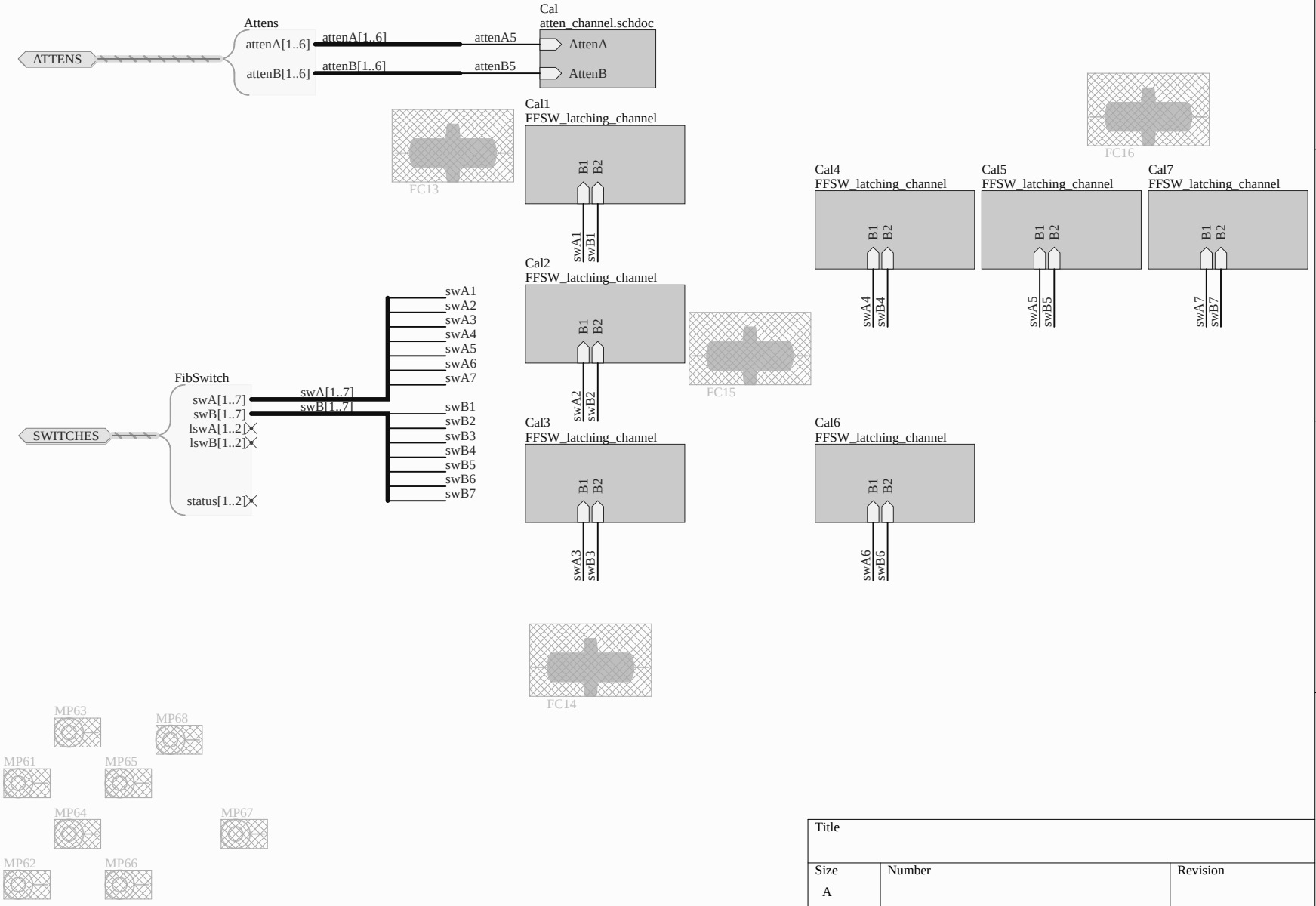




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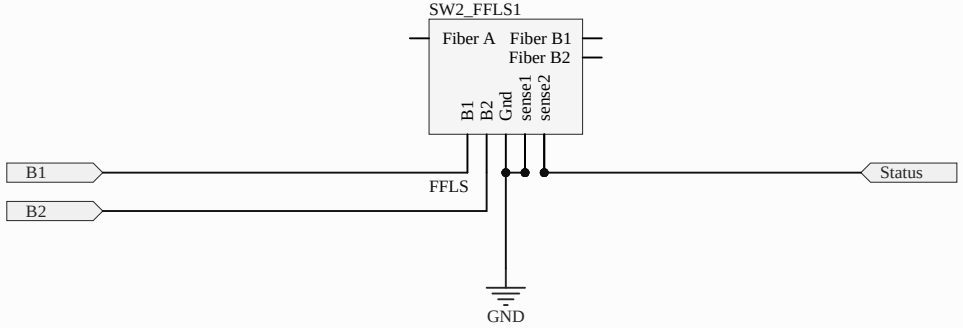
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PROJECT REVISION:	DOCUMENT REVISION:	DESIGN ITEM:	
TITLE: *			
SIZE: A		DWG NO.: Atten_Drivers.SchDoc	REV
SCALE:	FILE NAME: Atten_Drivers.SchDoc	SHEET: 8	OF: 16



Title		
Size A	Number	Revision
Date: 10/21/2026		Sheet of
File: C:\Users\..\CAL_Switching.SchDoc		Drawn By:

Pins are 0.5x0.5 x 2.8mm long FFLC. 1x7 unspecified pitch (looks 0.1in)
MillMax 801-41-007-10-012000

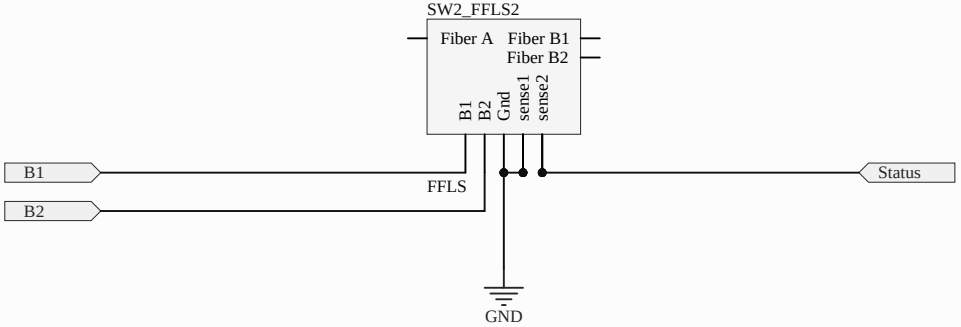
~125Ohm, 40mA @ 5V
SW2_FFLS1



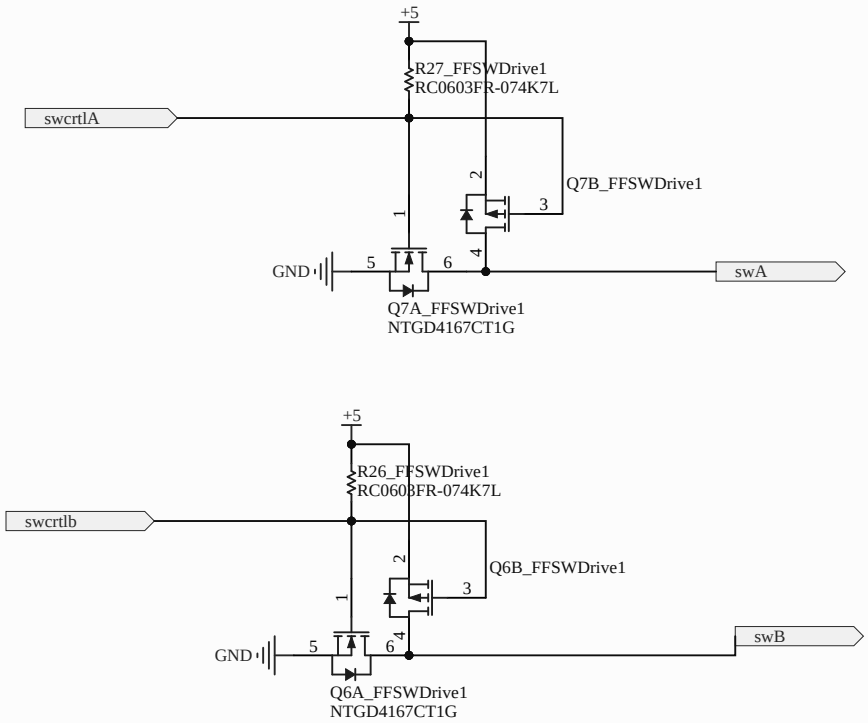
Title		
Size: A4	Number	Revision
Date: 10/21/2026		Sheet of
File: C:\Users\..\FFLS_latching_channel.SchDoc		Drawn By:

Pins are 0.5x0.5 x 2.8mm long FFLC. 1x7 unspecified pitch (looks 0.1in)
MillMax 801-41-007-10-012000

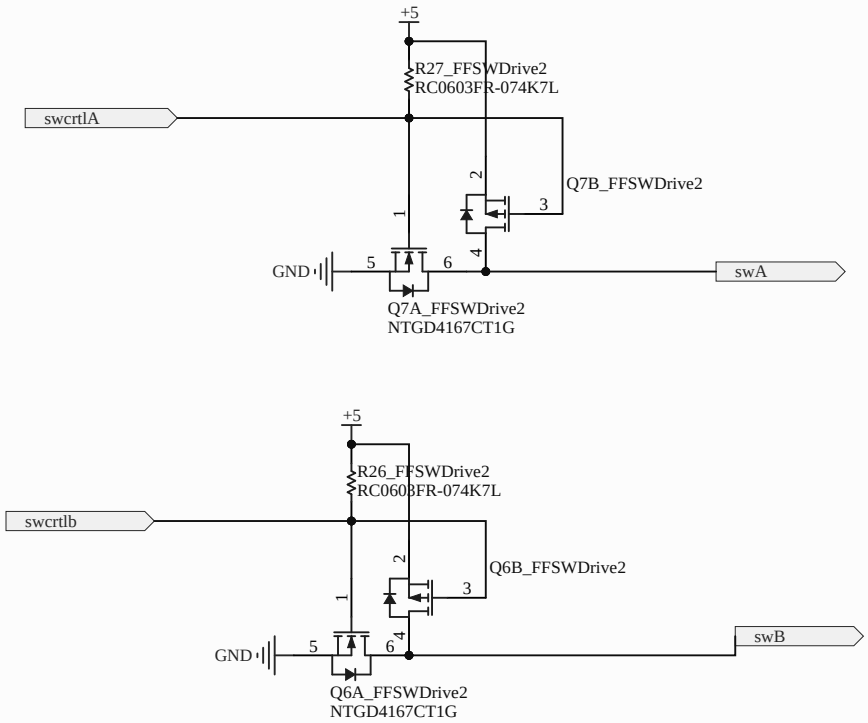
~125Ohm, 40mA @ 5V
SW2_FFLS2



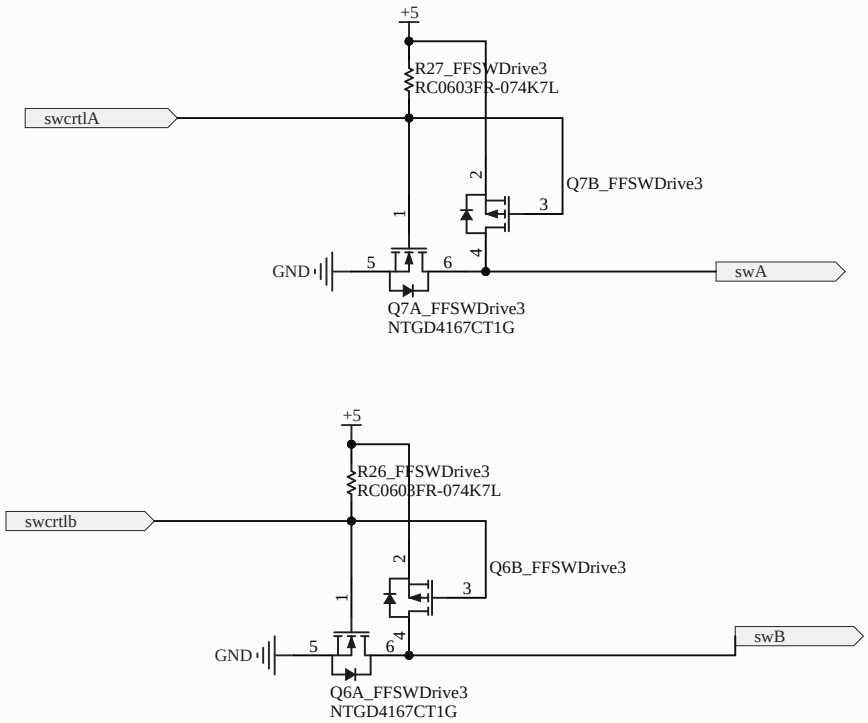
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Size: A4	Number	Revision
Date: 10/21/2026		Sheet of
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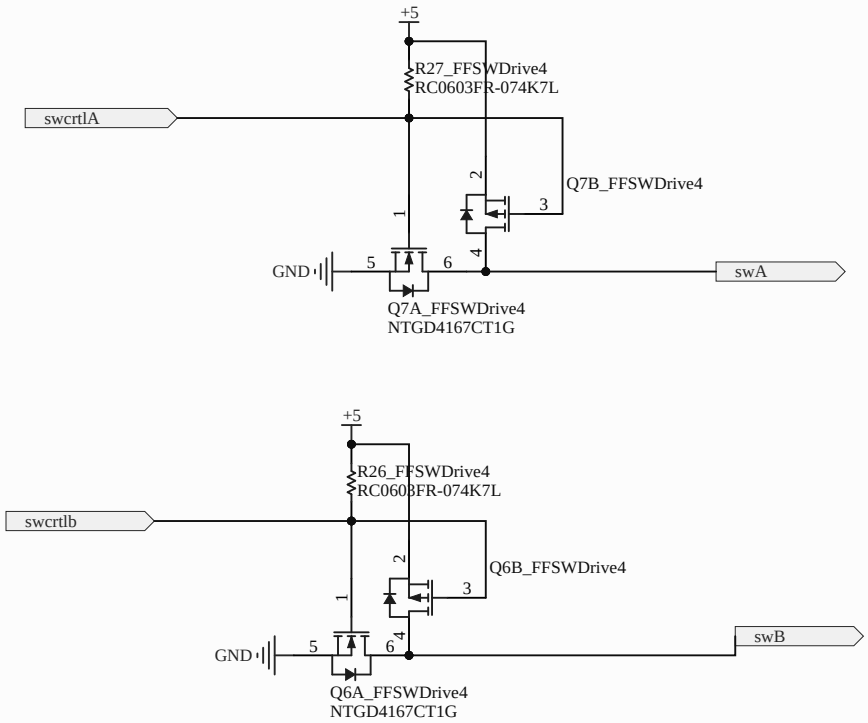
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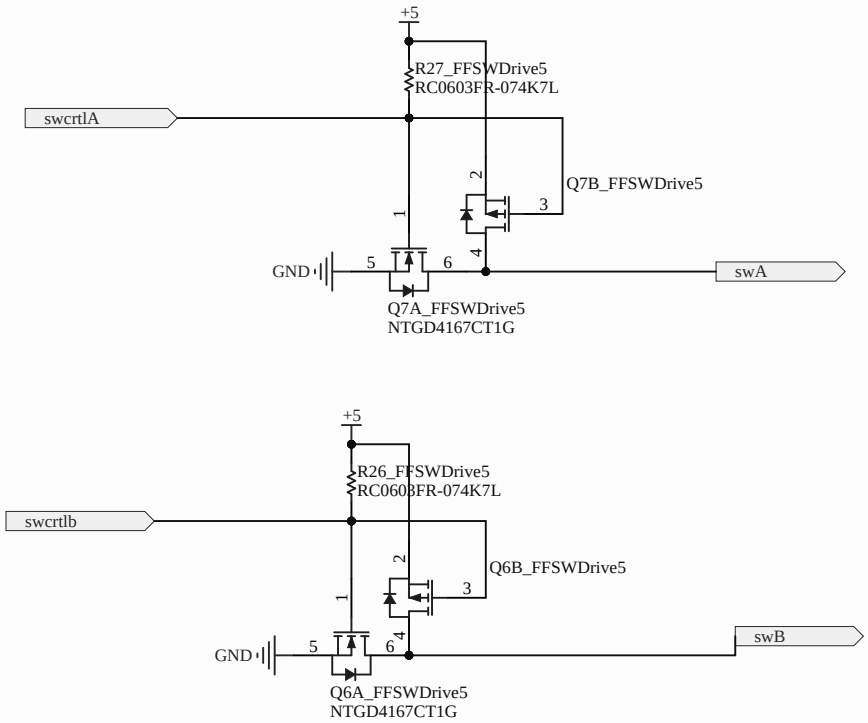
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Date: 10/21/2026		Sheet of
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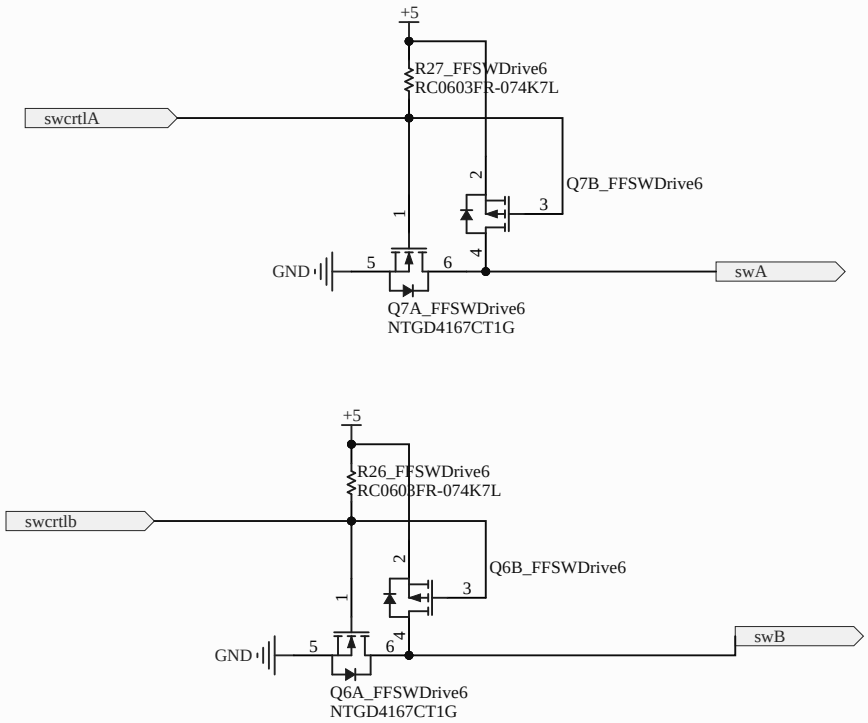
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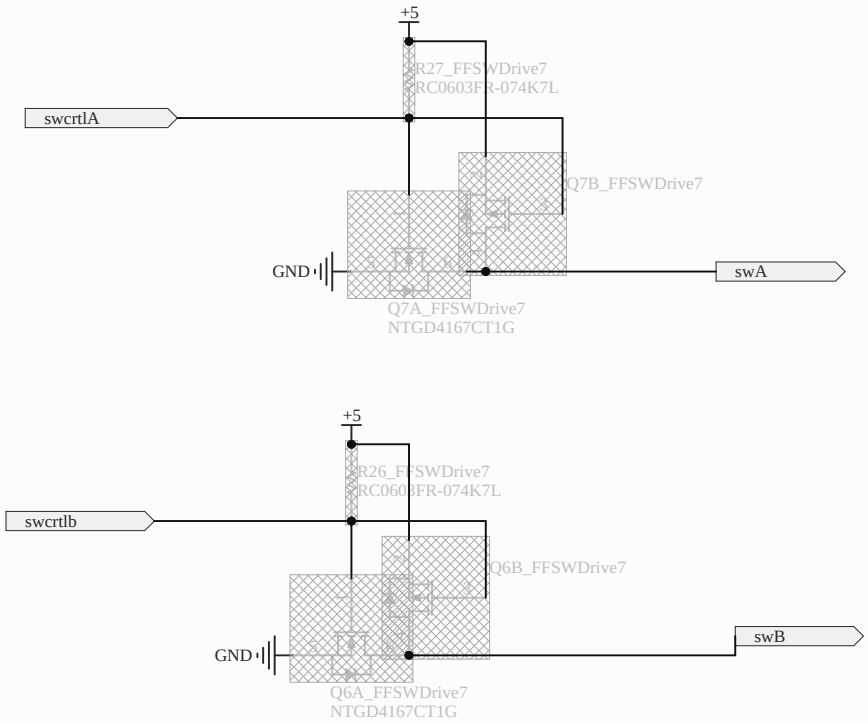
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Size A	Number		Revision
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Date: 10/21/2026		Sheet of
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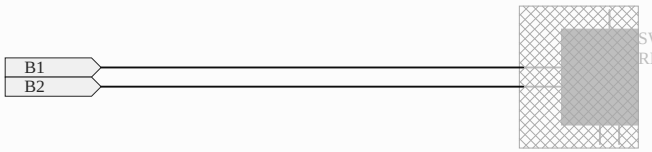


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Title		
Size A	Number	Revision
Date: 10/21/2026		Sheet of
File: C:\Users\..\FFSW_Drive.SchDoc		Drawn By:

REVISION	DESCRIPTION	DATE	APPROVED



4.5V @ 40mA ~180mW to switch

Pins are 0.5 x 2.9mm long FFSW
Pins are OD0.65mm x >3mm long FFSW-F
MillMax 0553-2-15-15-11-27-10-0 (Hex Pres fit)
MillMax 0552-2-15-15-11-27-10-0 (Solder)

MillMax Pin Sockets

J1_ASB1
0552-2-15-15-11-27-10-0

J3_ASB1
0552-2-15-15-11-27-10-0

J2_ASB1
0552-2-15-15-11-27-10-0

J4_ASB1
0552-2-15-15-11-27-10-0

APPROVALS		DATE
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CHK:	*	--/--/--

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FAB DWG:	<FAB DWG NO>
PCB DWG:	<PCB DWG NO>

PROJECT: *

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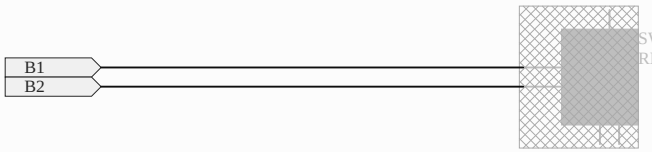
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SIZE	DWG NO.	REV
A	FFSW_latching_channel.SchDoc	

SCALE: | FILE NAME: FFSW_latching_channel.SchDoc | SHEET 11 OF 16

REVISION	DESCRIPTION	DATE	APPROVED



4.5V @ 40mA ~180mW to switch

Pins are 0.5 x 2.9mm long FFSW
Pins are OD0.65mm x >3mm long FFSW-F
MillMax 0553-2-15-15-11-27-10-0 (Hex Pres fit)
MillMax 0552-2-15-15-11-27-10-0 (Solder)

MillMax Pin Sockets

J1_ASB2
0552-2-15-15-11-27-10-0

J3_ASB2
0552-2-15-15-11-27-10-0

J2_ASB2
0552-2-15-15-11-27-10-0

J4_ASB2
0552-2-15-15-11-27-10-0

APPROVALS		DATE
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DSN:	*	--/--/--
CHK:	*	--/--/--

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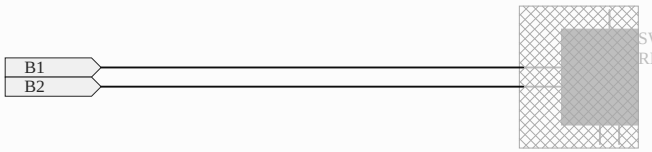
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TITLE: *

SIZE	DWG NO.	REV
A	FFSW_latching_channel.SchDoc	

SCALE: | FILE NAME: FFSW_latching_channel.SchDoc | SHEET 11 OF 16

REVISION	DESCRIPTION	DATE	APPROVED



4.5V @ 40mA ~180mW to switch

Pins are 0.5 x 2.9mm long FFSW
Pins are OD0.65mm x >3mm long FFSW-F
MillMax 0553-2-15-15-11-27-10-0 (Hex Pres fit)
MillMax 0552-2-15-15-11-27-10-0 (Solder)

MillMax Pin Sockets

J1_ASB3
0552-2-15-15-11-27-10-0

J3_ASB3
0552-2-15-15-11-27-10-0

J2_ASB3
0552-2-15-15-11-27-10-0

J4_ASB3
0552-2-15-15-11-27-10-0

APPROVALS		DATE
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DSN:	*	--/--/--
CHK:	*	--/--/--

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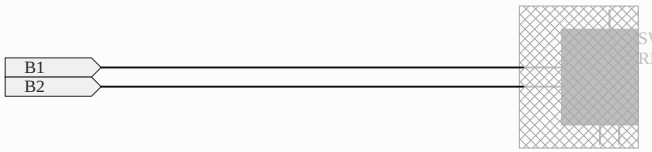
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TITLE: *

SIZE	DWG NO.	REV
A	FFSW_latching_channel.SchDoc	

SCALE: | FILE NAME: FFSW_latching_channel.SchDoc | SHEET 11 OF 16

REVISION	DESCRIPTION	DATE	APPROVED



4.5V @ 40mA ~180mW to switch

Pins are 0.5 x 2.9mm long FFSW
Pins are OD0.65mm x >3mm long FFSW-F
MillMax 0553-2-15-15-11-27-10-0 (Hex Pres fit)
MillMax 0552-2-15-15-11-27-10-0 (Solder)

MillMax Pin Sockets

J1_ASR1
0552-2-15-15-11-27-10-0

J3_ASR1
0552-2-15-15-11-27-10-0

J2_ASR1
0552-2-15-15-11-27-10-0

J4_ASR1
0552-2-15-15-11-27-10-0

APPROVALS		DATE
ENG:	*	--/--/--
DSN:	*	--/--/--
CHK:	*	--/--/--

REFERENCE DOCUMENTS	
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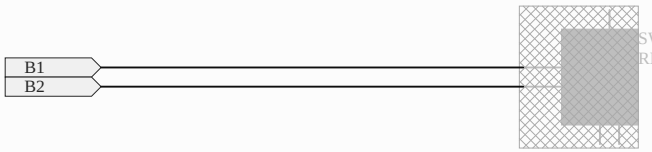
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SIZE	DWG NO.	REV
A	FFSW_latching_channel.SchDoc	

SCALE: | FILE NAME: FFSW_latching_channel.SchDoc | SHEET 11 OF 16

REVISION	DESCRIPTION	DATE	APPROVED



4.5V @ 40mA ~180mW to switch

Pins are 0.5 x 2.9mm long FFSW
Pins are OD0.65mm x >3mm long FFSW-F
MillMax 0553-2-15-15-11-27-10-0 (Hex Pres fit)
MillMax 0552-2-15-15-11-27-10-0 (Solder)

MillMax Pin Sockets

J1_ASR2
0552-2-15-15-11-27-10-0

J3_ASR2
0552-2-15-15-11-27-10-0

J2_ASR2
0552-2-15-15-11-27-10-0

J4_ASR2
0552-2-15-15-11-27-10-0

APPROVALS		DATE
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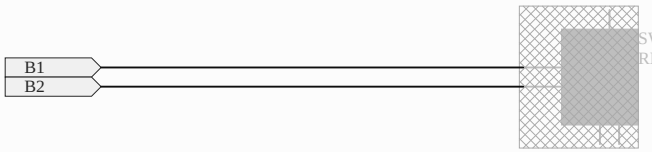
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SIZE	DWG NO.	REV
A	FFSW_latching_channel.SchDoc	

SCALE: | FILE NAME: FFSW_latching_channel.SchDoc | SHEET 11 OF 16

REVISION	DESCRIPTION	DATE	APPROVED



4.5V @ 40mA ~180mW to switch

Pins are 0.5 x 2.9mm long FFSW
Pins are OD0.65mm x >3mm long FFSW-F
MillMax 0553-2-15-15-11-27-10-0 (Hex Pres fit)
MillMax 0552-2-15-15-11-27-10-0 (Solder)

MillMax Pin Sockets

J1_ASR3
0552-2-15-15-11-27-10-0

J3_ASR3
0552-2-15-15-11-27-10-0

J2_ASR3
0552-2-15-15-11-27-10-0

J4_ASR3
0552-2-15-15-11-27-10-0

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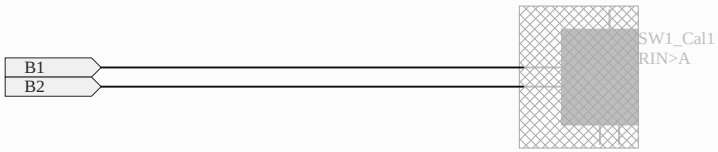
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SCALE: | FILE NAME: FFSW_latching_channel.SchDoc | SHEET 11 OF 16

REVISION	DESCRIPTION	DATE	APPROVED



4.5V @ 40mA ~180mW to switch

Pins are 0.5 x 2.9mm long FFSW
Pins are OD0.65mm x >3mm long FFSW-F
MillMax 0553-2-15-15-11-27-10-0 (Hex Pres fit)
MillMax 0552-2-15-15-11-27-10-0 (Solder)

MillMax Pin Sockets

J1_Cal1
0552-2-15-15-11-27-10-0

J3_Cal1
0552-2-15-15-11-27-10-0

J2_Cal1
0552-2-15-15-11-27-10-0

J4_Cal1
0552-2-15-15-11-27-10-0

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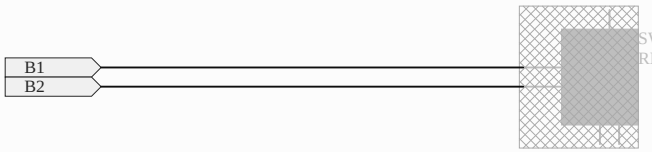
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SCALE: | FILE NAME: FFSW_latching_channel.SchDoc | SHEET 11 OF 16

REVISION	DESCRIPTION	DATE	APPROVED



4.5V @ 40mA ~180mW to switch

Pins are 0.5 x 2.9mm long FFSW
Pins are OD0.65mm x >3mm long FFSW-F
MillMax 0553-2-15-15-11-27-10-0 (Hex Pres fit)
MillMax 0552-2-15-15-11-27-10-0 (Solder)

MillMax Pin Sockets

J1_Cal2
0552-2-15-15-11-27-10-0

J3_Cal2
0552-2-15-15-11-27-10-0

J2_Cal2
0552-2-15-15-11-27-10-0

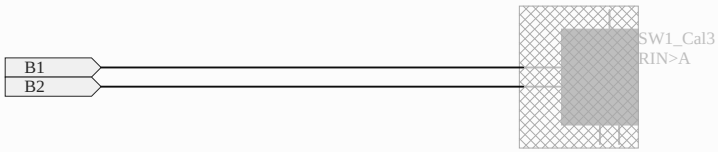
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SCALE:	FILE NAME: FFSW_latching_channel.SchDoc	SHEET: 11 OF 16	

REVISION	DESCRIPTION	DATE	APPROVED



4.5V @ 40mA ~180mW to switch

Pins are 0.5 x 2.9mm long FFSW
Pins are OD0.65mm x >3mm long FFSW-F
MillMax 0553-2-15-15-11-27-10-0 (Hex Pres fit)
MillMax 0552-2-15-15-11-27-10-0 (Solder)

MillMax Pin Sockets

J1_Cal3
0552-2-15-15-11-27-10-0

J3_Cal3
0552-2-15-15-11-27-10-0

J2_Cal3
0552-2-15-15-11-27-10-0

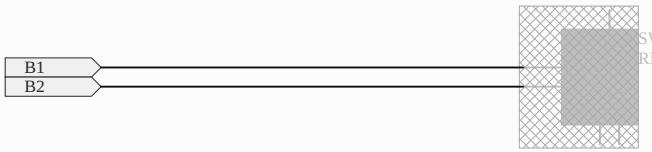
J4_Cal3
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SCALE:	FILE NAME: FFSW_latching_channel.SchDoc		SHEET: 11 OF 16

REVISION	DESCRIPTION	DATE	APPROVED



4.5V @ 40mA ~180mW to switch

Pins are 0.5 x 2.9mm long FFSW
Pins are OD0.65mm x >3mm long FFSW-F
MillMax 0553-2-15-15-11-27-10-0 (Hex Pres fit)
MillMax 0552-2-15-15-11-27-10-0 (Solder)

MillMax Pin Sockets

J1_Cal4
0552-2-15-15-11-27-10-0

J3_Cal4
0552-2-15-15-11-27-10-0

J2_Cal4
0552-2-15-15-11-27-10-0

J4_Cal4
0552-2-15-15-11-27-10-0

APPROVALS		DATE
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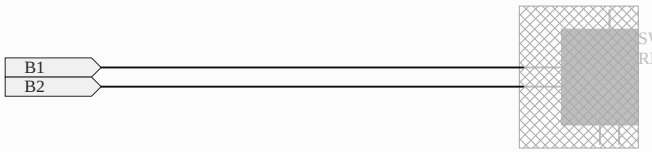
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SCALE: | FILE NAME: FFSW_latching_channel.SchDoc | SHEET 11 OF 16

REVISION	DESCRIPTION	DATE	APPROVED



4.5V @ 40mA ~180mW to switch

Pins are 0.5 x 2.9mm long FFSW
Pins are OD0.65mm x >3mm long FFSW-F
MillMax 0553-2-15-15-11-27-10-0 (Hex Pres fit)
MillMax 0552-2-15-15-11-27-10-0 (Solder)

MillMax Pin Sockets

J1_Cal5
0552-2-15-15-11-27-10-0

J3_Cal5
0552-2-15-15-11-27-10-0

J2_Cal5
0552-2-15-15-11-27-10-0

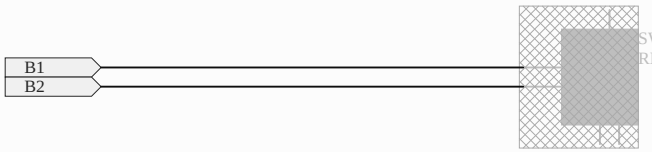
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SIZE: A	DWG NO.: FFSW_latching_channel.SchDoc		REV:
SCALE:	FILE NAME: FFSW_latching_channel.SchDoc	SHEET: 11	OF: 16

REVISION	DESCRIPTION	DATE	APPROVED



4.5V @ 40mA ~180mW to switch

Pins are 0.5 x 2.9mm long FFSW
Pins are OD0.65mm x >3mm long FFSW-F
MillMax 0553-2-15-15-11-27-10-0 (Hex Pres fit)
MillMax 0552-2-15-15-11-27-10-0 (Solder)

MillMax Pin Sockets

J1_Cal6
0552-2-15-15-11-27-10-0

J3_Cal6
0552-2-15-15-11-27-10-0

J2_Cal6
0552-2-15-15-11-27-10-0

J4_Cal6
0552-2-15-15-11-27-10-0

APPROVALS		DATE
ENG:	*	--/--/--
DSN:	*	--/--/--
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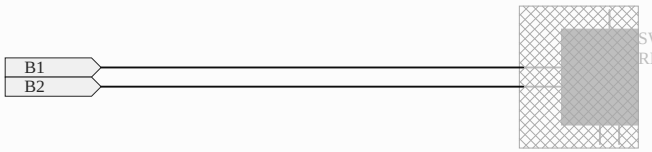
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SIZE	DWG NO.	REV
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SCALE: | FILE NAME: FFSW_latching_channel.SchDoc | SHEET 11 OF 16

REVISION	DESCRIPTION	DATE	APPROVED



4.5V @ 40mA ~180mW to switch

Pins are 0.5 x 2.9mm long FFSW
Pins are OD0.65mm x >3mm long FFSW-F
MillMax 0553-2-15-15-11-27-10-0 (Hex Pres fit)
MillMax 0552-2-15-15-11-27-10-0 (Solder)

MillMax Pin Sockets

J1_Cal7
0552-2-15-15-11-27-10-0

J3_Cal7
0552-2-15-15-11-27-10-0

J2_Cal7
0552-2-15-15-11-27-10-0

J4_Cal7
0552-2-15-15-11-27-10-0

APPROVALS		DATE
ENG:	*	--/--/--
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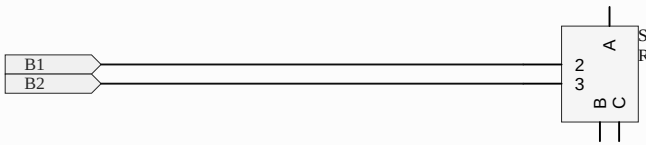
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SCALE: | FILE NAME: FFSW_latching_channel.SchDoc | SHEET 11 OF 16

REVISION	DESCRIPTION	DATE	APPROVED



SW1_TIBB1
RIN>A

4.5V @ 40mA ~180mW to switch

Pins are 0.5 x 2.9mm long FFSW
Pins are OD0.65mm x >3mm long FFSW-F
MillMax 0553-2-15-15-11-27-10-0 (Hex Pres fit)
MillMax 0552-2-15-15-11-27-10-0 (Solder)

MillMax Pin Sockets

J1_TIBB1
0552-2-15-15-11-27-10-0

J3_TIBB1
0552-2-15-15-11-27-10-0

J2_TIBB1
0552-2-15-15-11-27-10-0

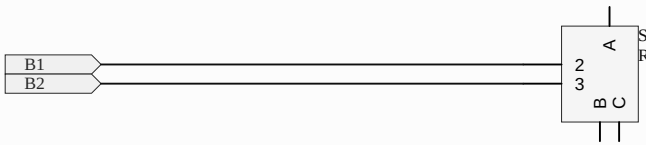
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SCALE:	FILE NAME: FFSW_latching_channel.SchDoc	SHEET: 11	OF: 16

REVISION	DESCRIPTION	DATE	APPROVED



SW1_TIBB2
RIN>A

4.5V @ 40mA ~180mW to switch

Pins are 0.5 x 2.9mm long FFSW
Pins are OD0.65mm x >3mm long FFSW-F
MillMax 0553-2-15-15-11-27-10-0 (Hex Pres fit)
MillMax 0552-2-15-15-11-27-10-0 (Solder)

MillMax Pin Sockets

J1_TIBB2
0552-2-15-15-11-27-10-0

J3_TIBB2
0552-2-15-15-11-27-10-0

J2_TIBB2
0552-2-15-15-11-27-10-0

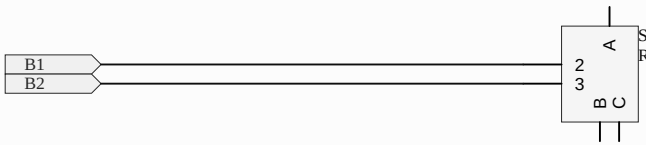
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SCALE:	FILE NAME: FFSW_latching_channel.SchDoc	SHEET: 11	OF: 16

REVISION	DESCRIPTION	DATE	APPROVED



SW1_TIBB3
RIN>A

4.5V @ 40mA ~180mW to switch

Pins are 0.5 x 2.9mm long FFSW
Pins are OD0.65mm x >3mm long FFSW-F
MillMax 0553-2-15-15-11-27-10-0 (Hex Pres fit)
MillMax 0552-2-15-15-11-27-10-0 (Solder)

MillMax Pin Sockets

J1_TIBB3
0552-2-15-15-11-27-10-0

J3_TIBB3
0552-2-15-15-11-27-10-0

J2_TIBB3
0552-2-15-15-11-27-10-0

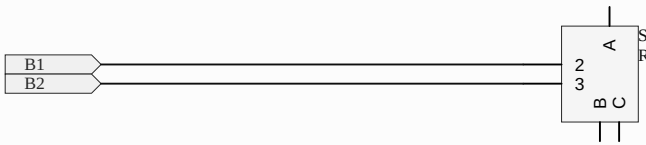
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SCALE:	FILE NAME: FFSW_latching_channel.SchDoc	SHEET: 11 OF 16	

REVISION	DESCRIPTION	DATE	APPROVED



SW1_TIBR1
RIN>A

4.5V @ 40mA ~180mW to switch

Pins are 0.5 x 2.9mm long FFSW
Pins are OD0.65mm x >3mm long FFSW-F
MillMax 0553-2-15-15-11-27-10-0 (Hex Pres fit)
MillMax 0552-2-15-15-11-27-10-0 (Solder)

MillMax Pin Sockets

J1_TIBR1
0552-2-15-15-11-27-10-0

J3_TIBR1
0552-2-15-15-11-27-10-0

J2_TIBR1
0552-2-15-15-11-27-10-0

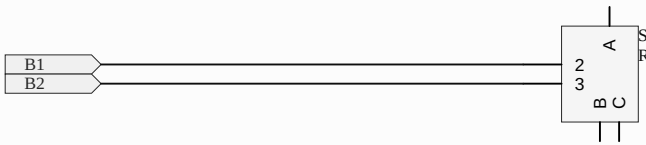
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REVISION	DESCRIPTION	DATE	APPROVED



SW1_TIBR2
RIN>A

4.5V @ 40mA ~180mW to switch

Pins are 0.5 x 2.9mm long FFSW
Pins are OD0.65mm x >3mm long FFSW-F
MillMax 0553-2-15-15-11-27-10-0 (Hex Pres fit)
MillMax 0552-2-15-15-11-27-10-0 (Solder)

MillMax Pin Sockets

J1_TIBR2
0552-2-15-15-11-27-10-0

J3_TIBR2
0552-2-15-15-11-27-10-0

J2_TIBR2
0552-2-15-15-11-27-10-0

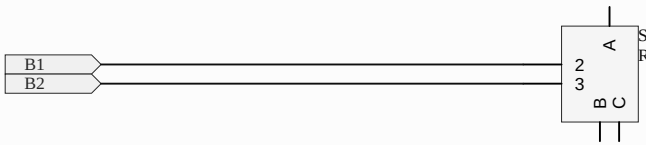
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PROJECT: *	Caltech - COO		
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SIZE: A		DWG NO.: FFSW_latching_channel.SchDoc	REV:
SCALE:	FILE NAME: FFSW_latching_channel.SchDoc	SHEET: 11 OF 16	

REVISION	DESCRIPTION	DATE	APPROVED



SW1_TIBR3
RIN>A

4.5V @ 40mA ~180mW to switch

Pins are 0.5 x 2.9mm long FFSW
Pins are OD0.65mm x >3mm long FFSW-F
MillMax 0553-2-15-15-11-27-10-0 (Hex Pres fit)
MillMax 0552-2-15-15-11-27-10-0 (Solder)

MillMax Pin Sockets

J1_TIBR3
0552-2-15-15-11-27-10-0

J3_TIBR3
0552-2-15-15-11-27-10-0

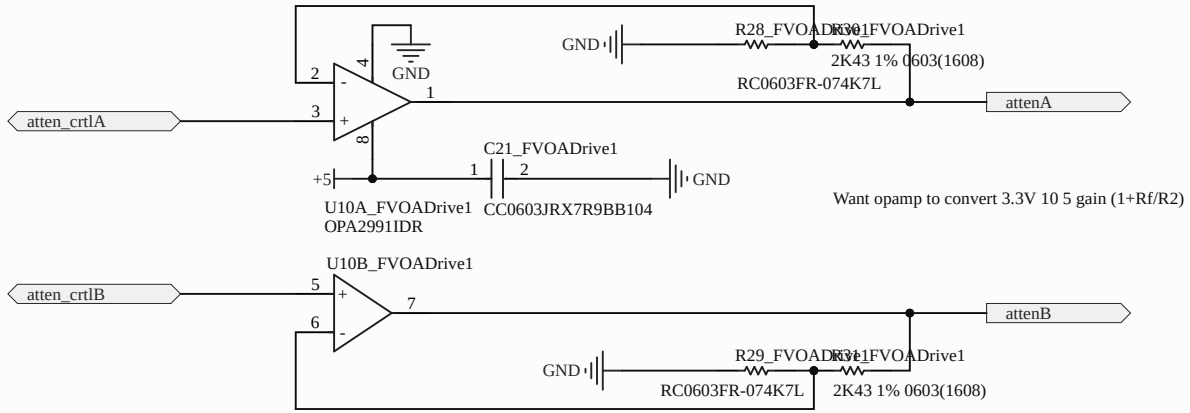
J2_TIBR3
0552-2-15-15-11-27-10-0

J4_TIBR3
0552-2-15-15-11-27-10-0

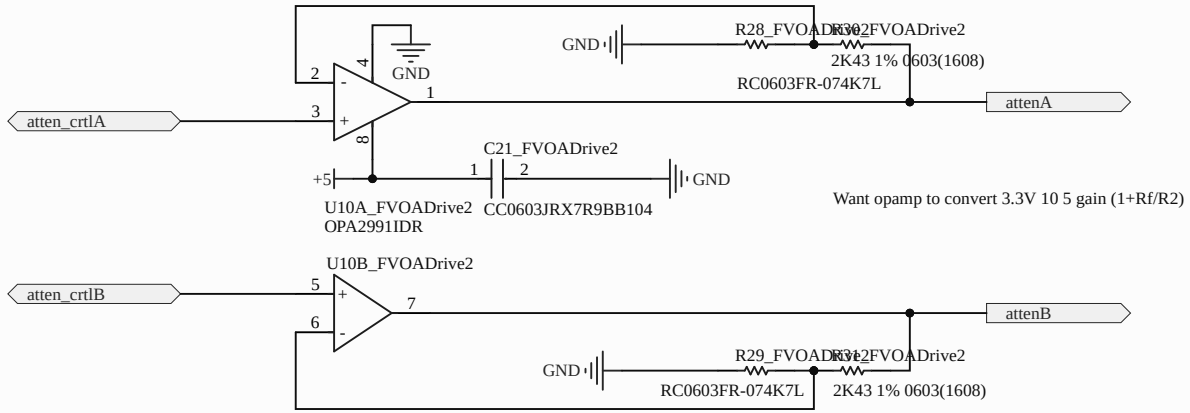
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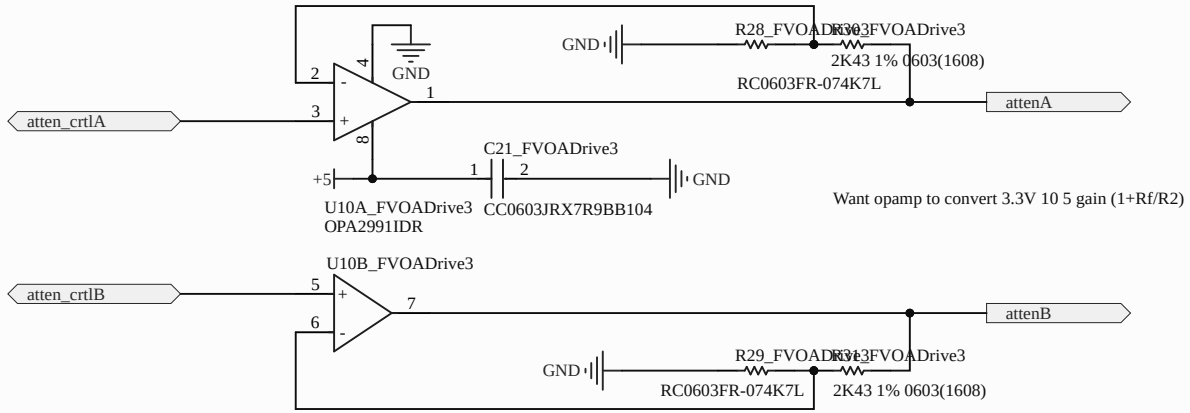
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SCALE:	FILE NAME: FFSW_latching_channel.SchDoc	SHEET: 11 OF 16	



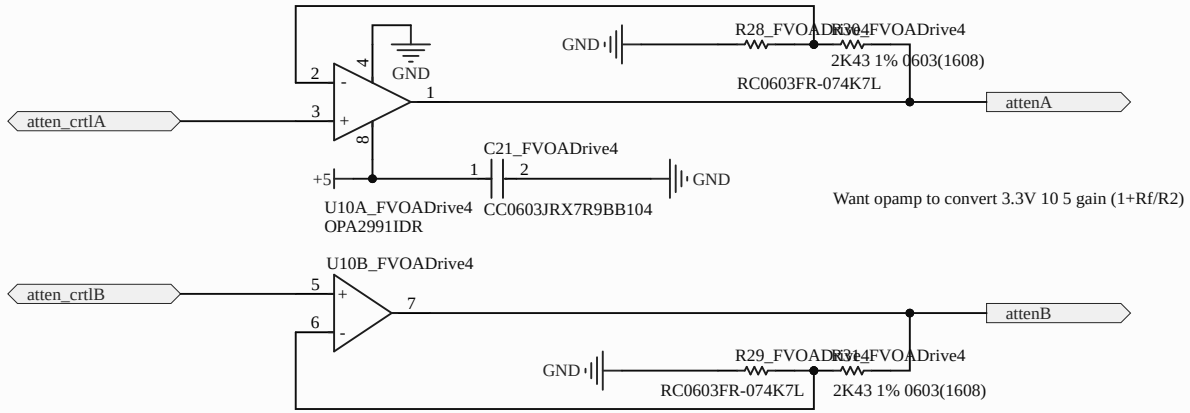
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Date: 10/21/2026		Sheet of
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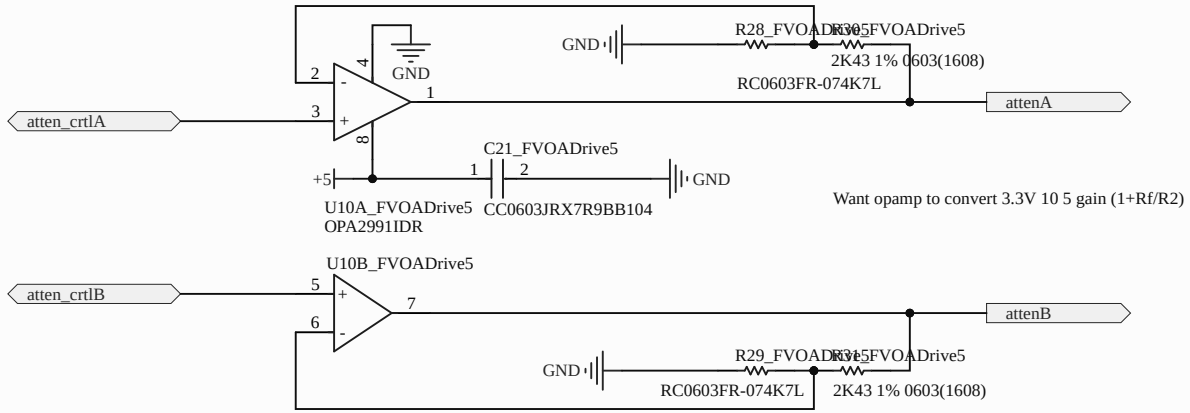
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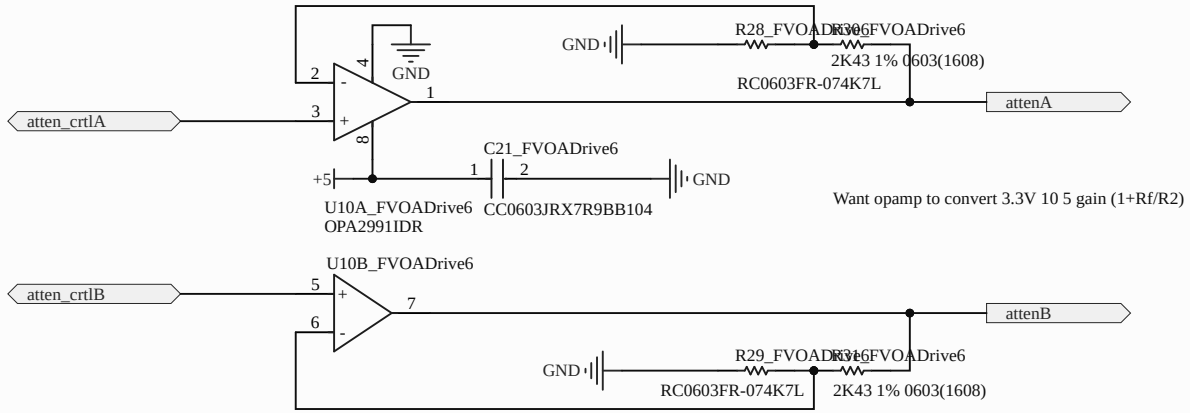
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File:	C:\Users\..\FVOA_Drive.SchDoc	Drawn By:	



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Date: 10/21/2026		Sheet of
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Size: A	Number	Revision
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REVISION	DESCRIPTION	DATE	APPROVED

AttenDrive
Atten_Drivers.SchDoc

SwitchDrive
Switch_Drivers.SchDoc

FibSwitch

RC0603FR-074K7L
U1
DS18B20Z+

Ctrl
Nucleo_H563.SchDoc

RC0603FR-074K7L

Power
Power.SchDoc

I'd like to have used: https://www.waveshare.com/ttl-to-rs485-c.htm

C1
CC0603JRX7R9BB104

R4
FC0603E50R0BTBST1

U2
THVD1429DT

J5
43650-0310

J6
436500210

J7
43650-0510

Prefab cables 1m: 1451320210, 1451320310, 1451320510

APPROVALS	DATE
ENG: *	
DSN: *	
CHK: *	

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BOM:	<BOM DOC NO>
ASSY DWG:	<ASSY DWG NO>
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PROJECT: Caltech - COO

PROJECT REVISION: | DOCUMENT REVISION: | DESIGN ITEM:

TITLE: *

SIZE: A | DWG NO.: HISPEC FIB Switchboard Control.SchDoc | REV:

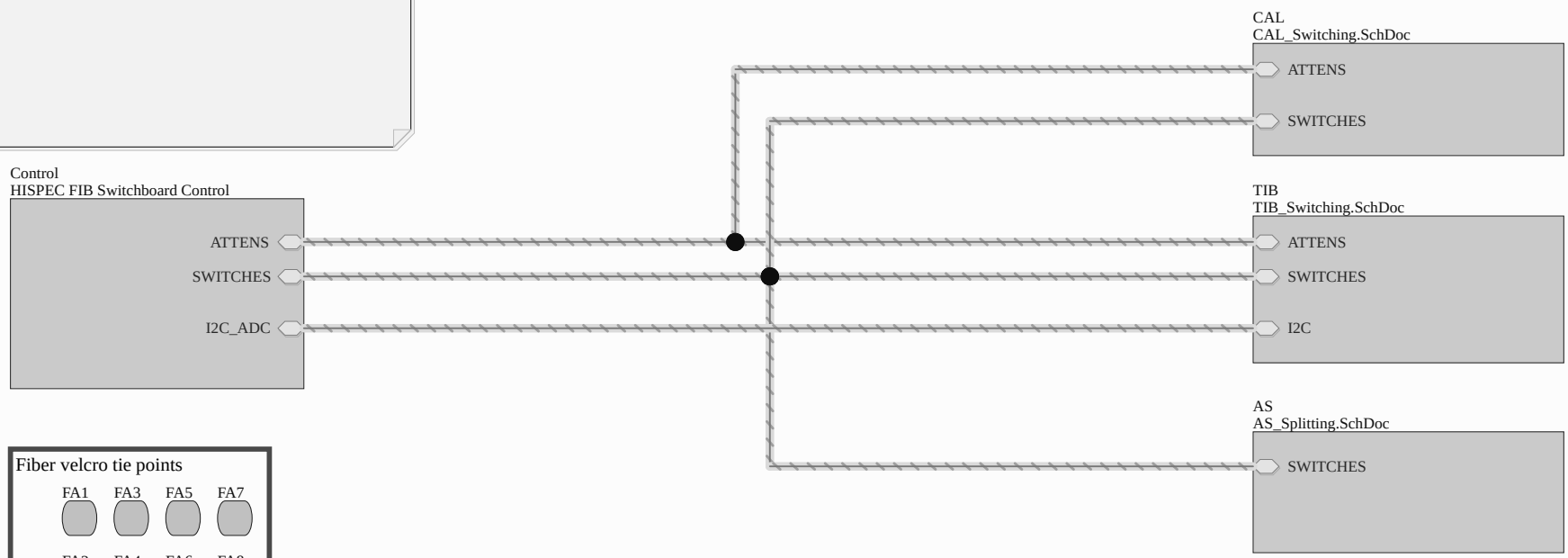
SCALE: | FILE NAME: HISPEC FIB Switchboard Control.SchDoc | SHEET 2 OF 16

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Connectors:
2 pin DC input 12-24 vdc, reverse protected TVS and reverse voltage protected.
5 pin to laser bank (RS485, 5VDC@3A max, GND, vsense)
3 pin power control connector (3v3 TTL to opto)

Power Circuitry:
12V to Nucleo @ 250mA max
Nucleo provides 5V (LD1117S50TR, 800mA max) and 3.3V (fed by by 5V LDO)

Notes:
- MEMS Switches are about 40mA pulses @ 5V
- Attenuators are ~9mA @ 5V full on.
- GPIO Expander and DACs operate on minimal power



Fiber velcro tie points

FA1 FA3 FA5 FA7

FA2 FA4 FA6 FA8

Screw Mounting Anchors

MP100	MP102	MP104	MP106	MP108
9774010482R	9774010482R	9774010482R	9774010482R	9774010482R
MP101	MP103	MP105	MP107	MP109
9774010482R	9774010482R	9774010482R	9774010482R	9774010482R

G1

Canno

COO Logo

APPROVALS	DATE
ENG: *	--/--/--
DSN: *	--/--/--
CHK: *	--/--/--

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BOM:	<BOM DOC NO>
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PROJECT: *

Caltech - COO

PROJECT REVISION: | DOCUMENT REVISION: | DESIGN ITEM:

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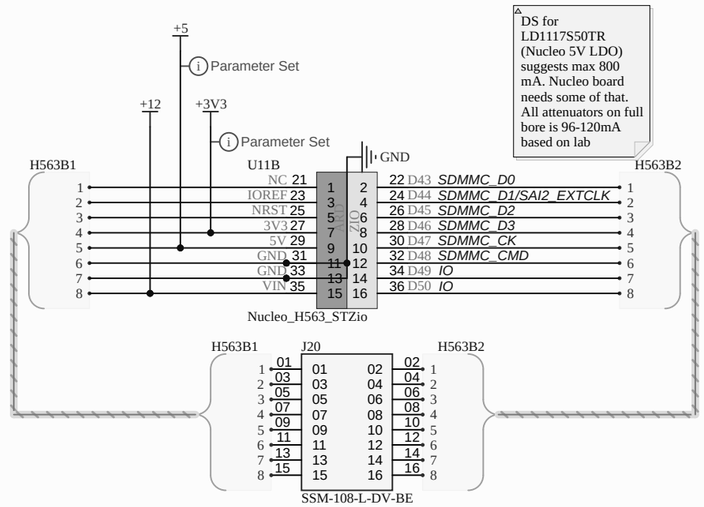
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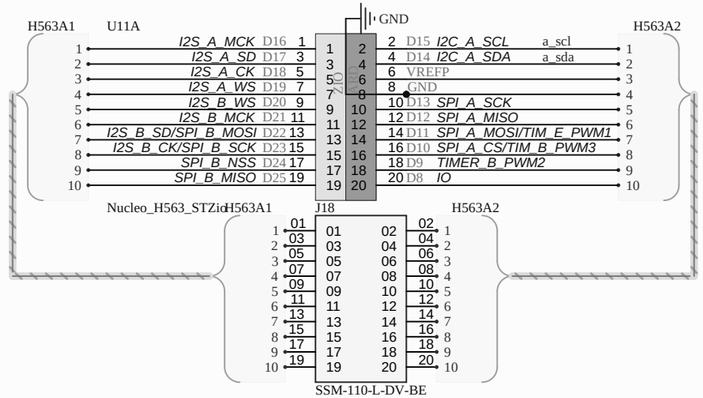
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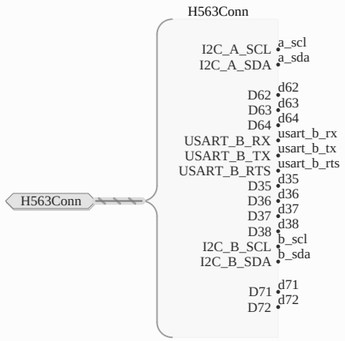
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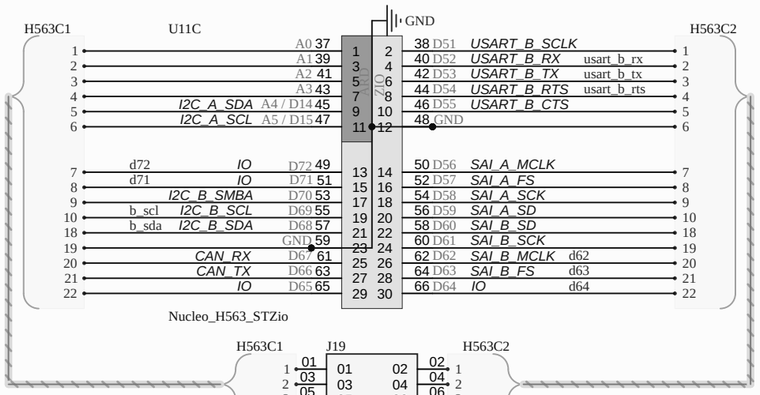
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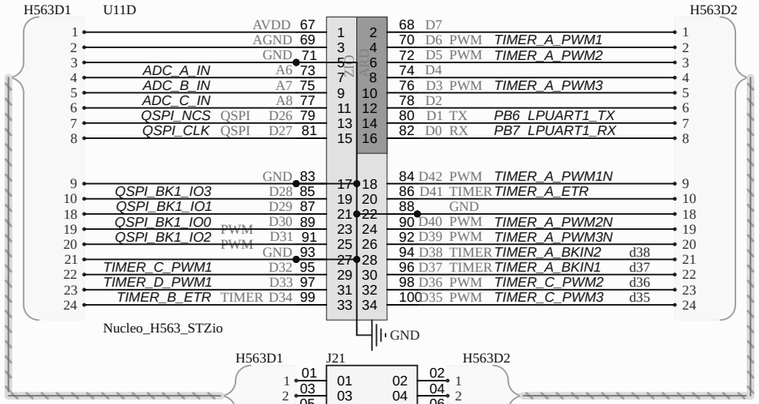
DS for LD1117S50TR (Nucleo 5V LDO) suggests max 800 mA. Nucleo board needs some of that. All attenuators on full bore is 96-120mA based on lab











CN10:
28:d38:tib
30:d37:calr
32:d36:calb
34:d35:as
CN7:
2:a_scl:adc
4:a_sck:adc
CN8: (power only)
CN9:
4: usart_b_rx:rs485
6: usart_b_tx:rs485
8: usart_b_rts:rs485
13:d72:laser_on
15:d71:pd_on
19:b_scl:gpio/dac
21:b_sck:gpio/dac
26:d62:lsw1_stat
28:d63:lsw2_stat
30:d64:1wire

APPROVALS	DATE
ENG: *	
DSN: *	
CHK: *	

REFERENCE DOCUMENTS

BOM:	<BOM DOC NO>
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FAB DWG:	<FAB DWG NO>
PCB DWG:	<PCB DWG NO>

PROJECT *

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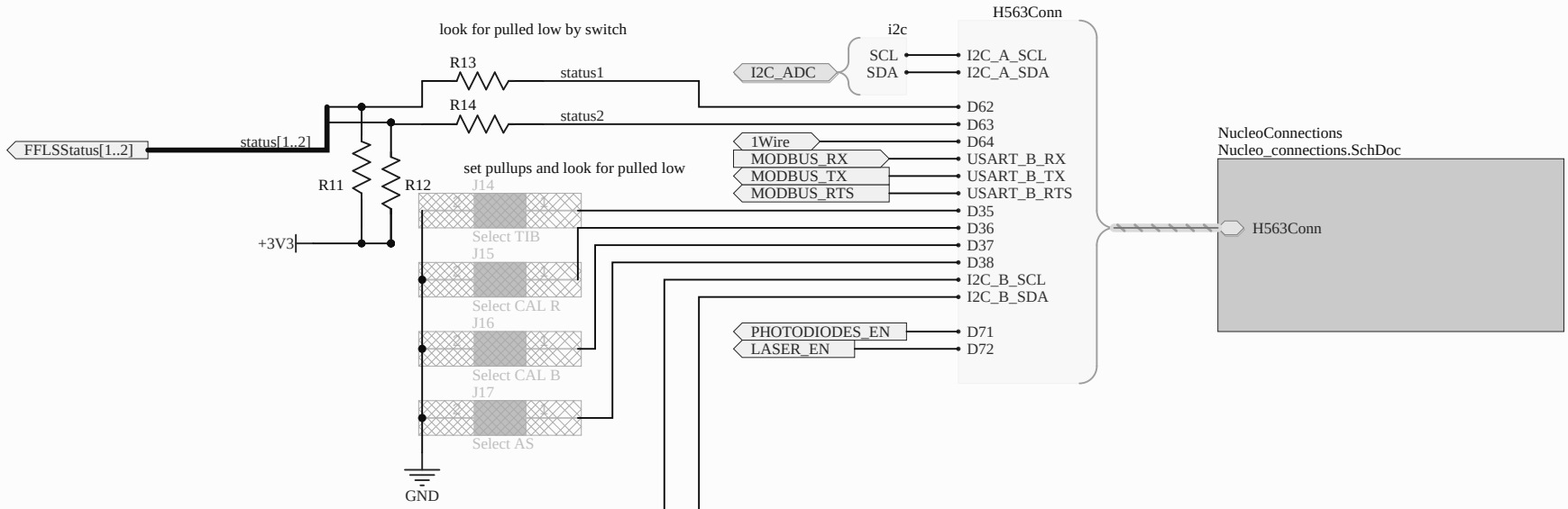
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SCALE: | FILE NAME Nucleo_connections.SchDoc | SHEET 4 OF 16

DWG NO <SCH DWG NO> REV | SHT 4

REVISION	DESCRIPTION	DATE	APPROVED



CN10:
28:d38:tib
30:d37:calr
32:d36:calb
34:d35:as
CN7:
2:a_scl:adc
4:a_sck:adc
CN8: (power only)
CN9:
4: usart_b_rx:rs485
6: usart_b_tx:rs485
8: usart_b_rts:rs485
13:d72:laser_on
15:d71:pd_on
19:b_scl:gpio/dac
21:b_sck:gpio/dac
26:d62:lsw1_stat
28:d63:lsw2_stat
30:d64:1wire

APPROVALS	DATE
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DSN: *	--/--/--
CHK: *	--/--/--

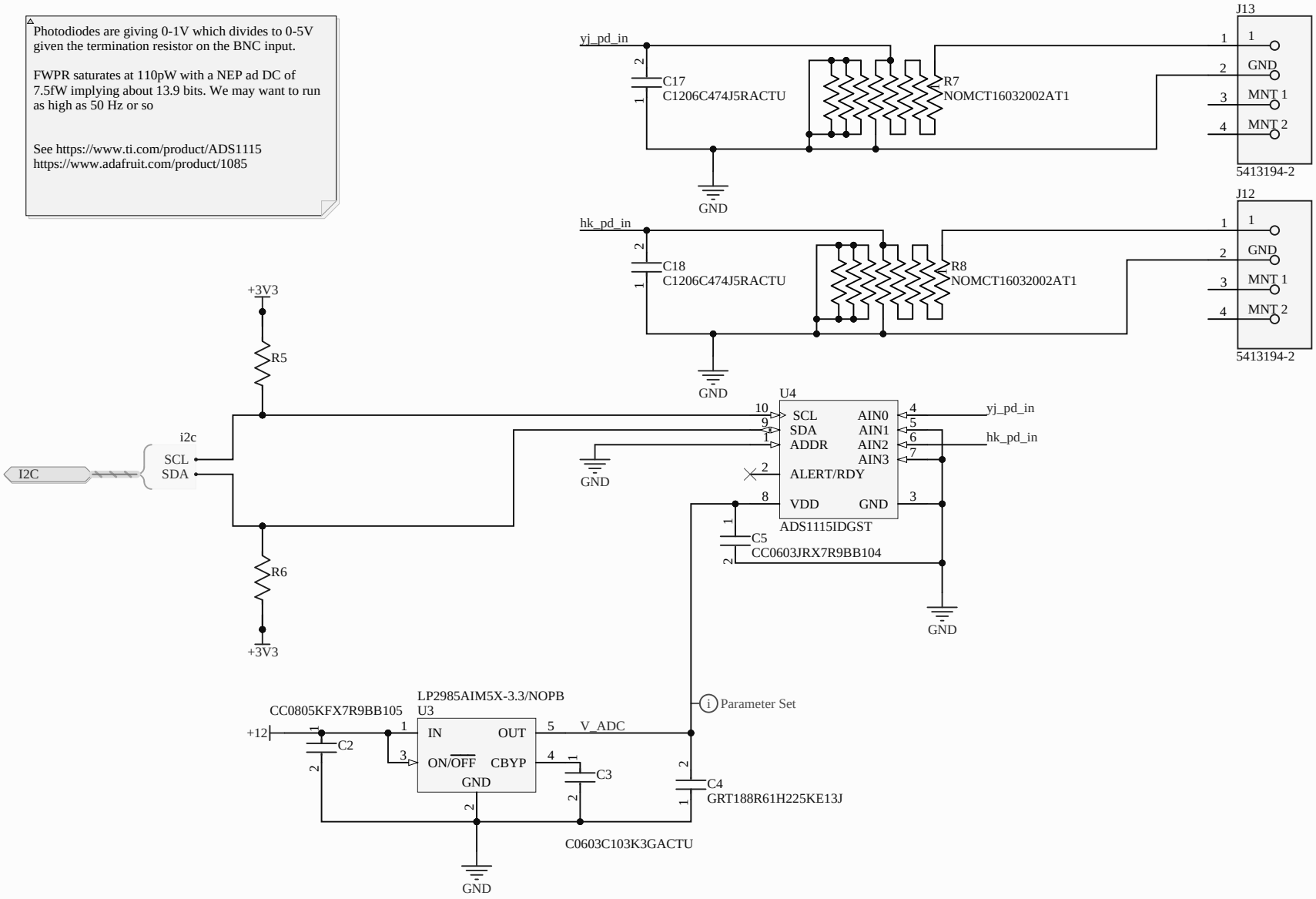
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PROJECT: *	Caltech - COO		
PROJECT REVISION:	DOCUMENT REVISION:	DESIGN ITEM:	
TITLE: *			
SIZE: A		DWG NO.: Nucleo_H563.SchDoc	REV:
SCALE:	FILE NAME: Nucleo_H563.SchDoc	SHEET: 3	OF: 16

Photodiodes are giving 0-1V which divides to 0-5V given the termination resistor on the BNC input.

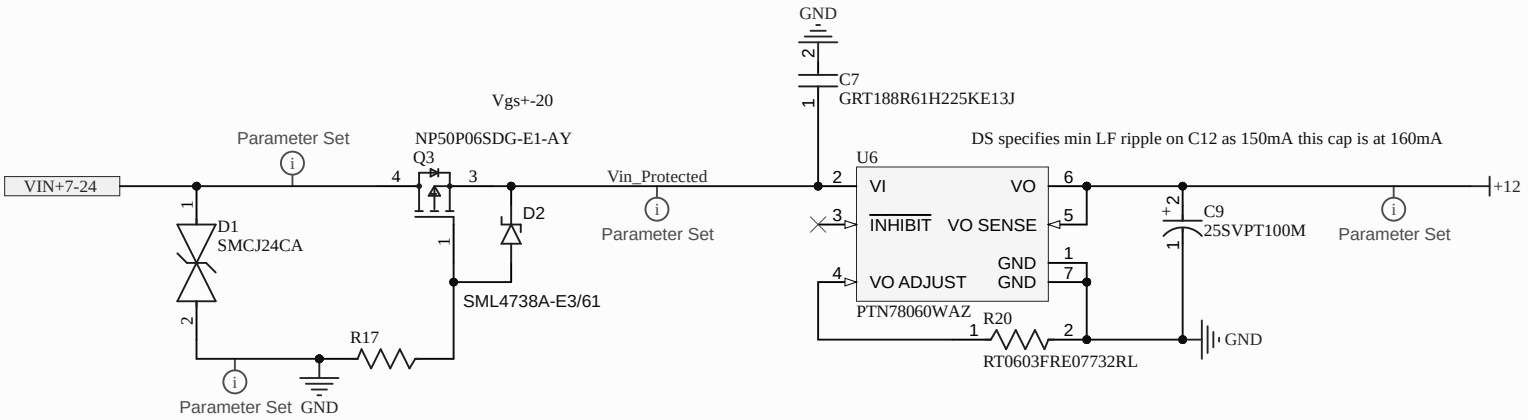
FWPR saturates at 110pW with a NEP ad DC of 7.5fW implying about 13.9 bits. We may want to run as high as 50 Hz or so

See https://www.ti.com/product/ADS1115
https://www.adafruit.com/product/1085



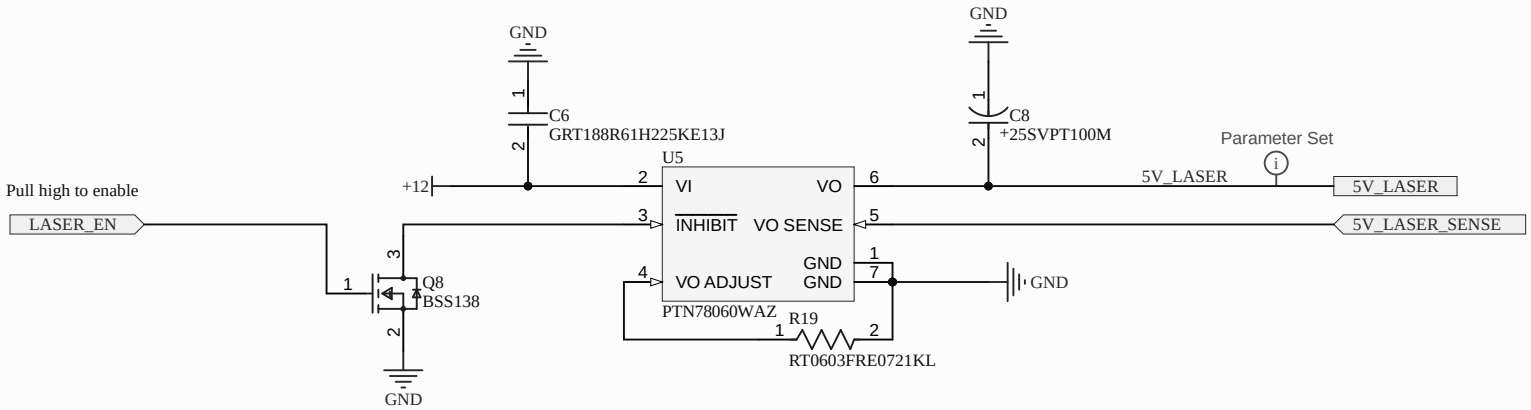
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REVISION	DESCRIPTION	DATE	APPROVED



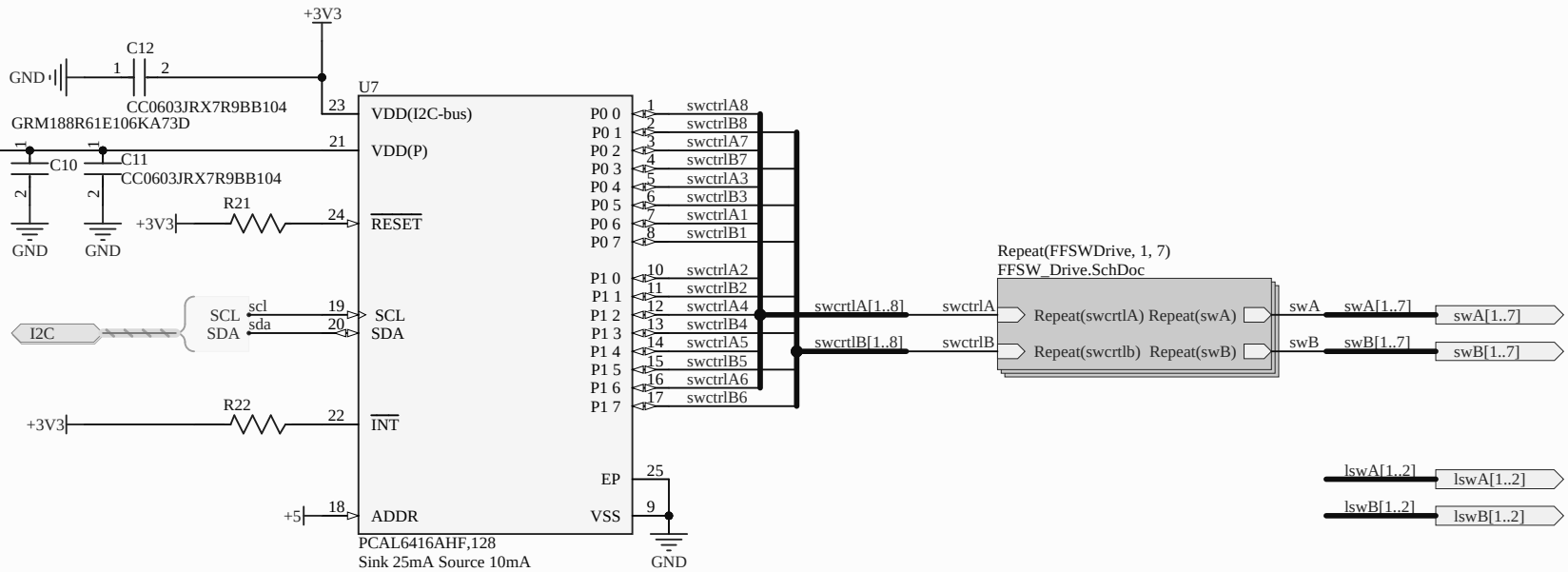
Modules have output current limit protection

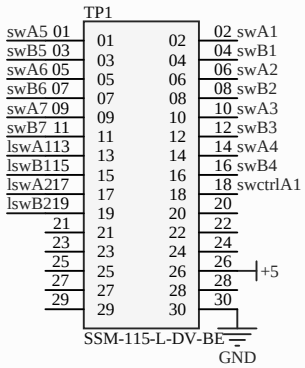
3V3 is sourced from Nucleo on that sheet.

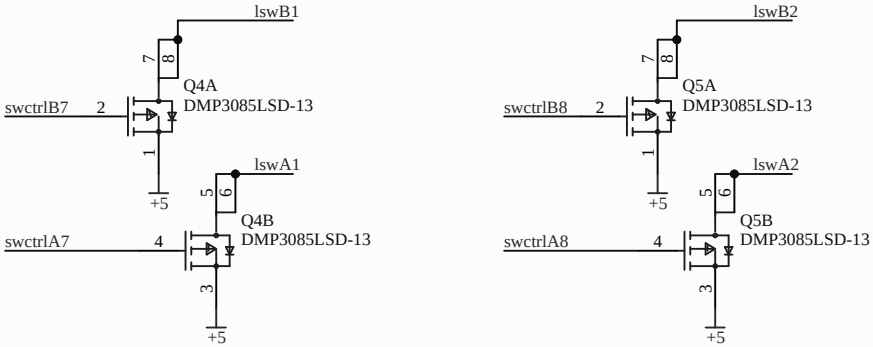


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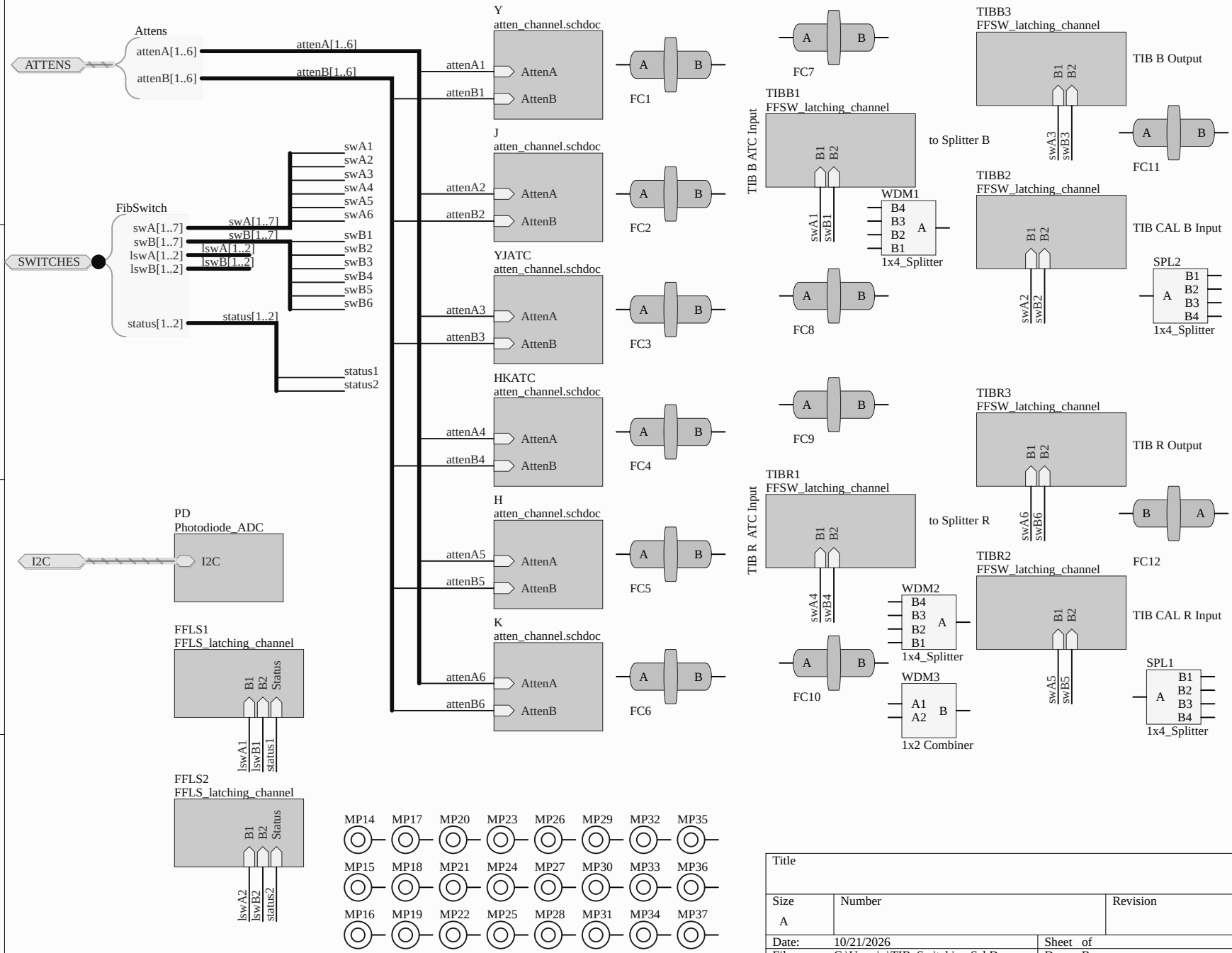
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SCALE:	FILE NAME Power.SchDoc	SHEET 5	OF 16







Title		
Size A	Number	Revision
Date: 10/21/2026		Sheet of
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Attens
ATTENS
attenA[1..6]
attenB[1..6]
attenA[1..6]
attenB[1..6]
FibSwitch
SWITCHES
swA[1..7]
swB[1..7]
lswA[1..2]
lswB[1..2]
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swB[1..7]
lswA[1..2]
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status[1..2]
swA1
swA2
swA3
swA4
swA5
swA6
swB1
swB2
swB3
swB4
swB5
swB6
status1
status2
Y
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J
atten_channel.schdoc
YJATC
atten_channel.schdoc
HKATC
atten_channel.schdoc
H
atten_channel.schdoc
K
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attenA1
attenB1
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attenB3
attenA4
attenB4
attenA5
attenB5
attenA6
attenB6
AttenA
AttenB
FC1
FC2
FC3
FC4
FC5
FC6
FC7
FC8
FC9
FC10
FC11
FC12
PD
Photodiode_ADC
I2C
I2C
FFLS1
FFLS_latching_channel
B1
B2
Status
lswA1
lswB1
status1
FFLS2
FFLS_latching_channel
B1
B2
Status
lswA2
lswB2
status2
TIBB1
FFSW_latching_channel
TIB B ATC Input
B1
B2
swA1
swB1
to Splitter B
WDM1
B4
B3
B2
B1
A
1x4_Splitter
TIBB3
FFSW_latching_channel
TIB B Output
B1
B2
swA3
swB3
TIBB2
FFSW_latching_channel
TIB CAL B Input
B1
B2
swA2
swB2
SPL2
A
B1
B2
B3
B4
1x4_Splitter
TIBR1
FFSW_latching_channel
TIB R ATC Input
B1
B2
swA4
swB4
to Splitter R
WDM2
B4
B3
B2
B1
A
1x4_Splitter
WDM3
A1
A2
B
1x2 Combiner
TIBR3
FFSW_latching_channel
TIB R Output
B1
B2
swA6
swB6
TIBR2
FFSW_latching_channel
TIB CAL R Input
B1
B2
swA5
swB5
SPL1
A
B1
B2
B3
B4
1x4_Splitter
MP14
MP17
MP20
MP23
MP26
MP29
MP32
MP35
MP15
MP18
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